

An Empirical Model of Large-Batch Training

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The motivation for batching is to reduce the number steps during training by smoothing the search space and increasing parallelism. This study provides a theoretical framework and experimental validation for an approach to determining an optimal batch size.

Analysis of the mathematical derivation and assumptions

The overall approach can be summarized as follows:

- 1) define the random variable for the gradient of a batch of a given batch size,
- 2) define the expectation and variance of this random variable,
- 3) approximate the loss improvement with the Taylor series expansion up to the second order derivative, given the gradient of the entire dataset at an optimization step, and through the minimization of this approximation determine the amount of the gradient that results in the optimal loss improvement,
- 4) approximate the expected loss improvement with the Taylor series expansion up to the second order derivative, given the gradient random variable for a batch at an optimization step, and through the minimization of this approximation of expectation determine the amount of the gradient that results in the optimal expected loss improvement.

θ is a parameter vector of dimensionality D . $L_x(\theta)$ is a scalar. $\nabla_\theta L(\theta)$, $G_{est}(\theta)$, and

$E_{x_{1...B} \sim \rho}[G_{est}(\theta)]$ are column vectors of dimensionality D . Different batch sizes result in

different random variables $G_{est}(\theta)$.

Equation 2.2.

$$\mathbb{E}_{x_1 \dots x_B \sim \rho}[G_{est}(\theta)] = G(\theta)$$

$$cov_{x_1 \dots x_B \sim \rho}[G_{est}(\theta)] = \frac{1}{B} \Sigma(\theta)$$

If $B = 1$ then the covariance matrix $cov_{x_1 \dots x_B \sim \rho}[G_{est}(\theta)]$ is equal to the per-example covariance matrix $\Sigma(\theta)$. If $B = \text{size of dataset}$, then each sampled value of $G_{est}(\theta)$ is a vector of the means in each dimension, and each entry in the covariance matrix $cov_{x_1 \dots x_B \sim \rho}[G_{est}(\theta)]$ is $(g_{est,i} - \mu_i)(g_{est,j} - \mu_j) = 0$.

The assumption is made that $cov_{x_1 \dots x_B \sim \rho}[G_{est}(\theta)]$ can be approximated at an arbitrary batch size B by dividing the per-example covariance matrix $\Sigma(\theta)$ by B . As B approaches the size of the dataset, the entries in $cov_{x_1 \dots x_B \sim \rho}[G_{est}(\theta)]$ approach 0.

There is no stated assumption about the diversity of the dataset. For any degree of data diversity, the entries in $cov_{x_1 \dots x_B \sim \rho}[G_{est}(\theta)]$ approach 0 as B approaches the size of the dataset. However, at lower data diversity, a smaller batch size may already capture the true dataset gradient. The later derived noise scale may capture the diversity of the dataset to an extent, as stated in the paper.

Equation 2.4.

$$L(\theta - \epsilon V) \approx L(\theta) - \epsilon G^T V + \frac{1}{2} \epsilon^2 V^T H V$$

The loss function was approximated around θ with the Taylor series expansion up to the second order derivative. This loss function is with respect to the true dataset gradient. V is a difference vector and ϵ is scaling factor. After substituting V with the gradient, ϵ is the learning rate (i.e. how much of the gradient is used to adjust each parameter).

We are looking for ϵ where the value of the approximation of the loss function is minimized and therefore the adjustment of θ is optimal. Equation 2.4 is a second degree polynomial. Taking its first derivative with respect to ϵ and setting it to zero provides ϵ_{max} . **The assumption is made that the Taylor series expansion up to the second order derivative accurately approximates the loss function for minimization purposes.** The purpose of the second order derivative term is to create a minimum, rather than to more accurately approximate the loss function in the vicinity of θ . This is an assumption about the shape of the true loss function, which may be dependent on the domain. Although a reference is made to the Taylor series expansion in section 2.4 (Assumptions and Caveats) of the paper, this assumption is not stated. The scaling laws are based on this assumption though the dependence on the critical batch size.

Equations 2.5 - 2.8.

$$\mathbb{E}[L(\theta - \epsilon G_{est})] \approx L(\theta) - \epsilon |G|^2 + \frac{1}{2} \epsilon^2 (G^T H G + \frac{tr(H \Sigma)}{B})$$

The batch size-specific random variables $G_{est}(\theta)$ are used. The loss function approximation is now a function of a random variable $G_{est}(\theta)$, defined by the batch size. The expectation of the loss function approximation is derived. The derivation of the last term is as follows.

$$G_{est}^T H G_{est} = tr(H G_{est} G_{est}^T)$$

$$\mathbb{E}[G_{est}G_{est}^T] = \mathbb{E}[G_{est}]\mathbb{E}[G_{est}^T] + \frac{1}{B}\Sigma$$

$$tr(H\mathbb{E}[G_{est}G_{est}^T]) = tr(H(\mathbb{E}[G_{est}]\mathbb{E}[G_{est}^T] + \frac{1}{B}\Sigma)) = tr(H(\mathbb{E}[G_{est}]\mathbb{E}[G_{est}^T])) + tr(\frac{1}{B}H\Sigma)$$

Use the expectation in equation 2.2.

$$tr(HGG^T) + tr(\frac{1}{B}H\Sigma) = G^THG + \frac{tr(H\Sigma)}{B}$$

By taking the first derivative of $\mathbb{E}[L(\theta - \epsilon G_{est})]$ with respect to ϵ and setting it to zero, the noise scale B_{noise} is derived.

$$\epsilon_{opt}(B) = \frac{|G|^2}{G^THG + \frac{tr(H\Sigma)}{B}} = \frac{\epsilon_{max}}{1 + \frac{1}{B} \frac{tr(H\Sigma)}{G^THG}} = \frac{\epsilon_{max}}{1 + \frac{1}{B}B_{noise}}$$

$$B_{noise} = \frac{tr(H\Sigma)}{G^THG}$$

The noise scale B_{noise} depends on the above assumptions. It depends on the assumption that the covariance matrix $cov_{x_1 \dots B \sim \rho}[G_{est}(\theta)]$ at an arbitrary batch size B can be approximated by dividing the per-example covariance matrix $\Sigma(\theta)$ by B . It also depends on the assumption that the Taylor series expansion up to the second order derivative accurately approximates the loss function for minimization purposes.

B_{noise} is the scalar ratio between i) the sum of the entries in the per-example gradient covariance matrix, scaled with Hessian for each parameter pair, and ii) the second order

derivative contribution in the Taylor series expansion of the loss function with the expected batch gradient as the difference vector, which is equal to the dataset gradient. B_{noise} does not depend on the batch size because the expected batch gradient is the same across batch sizes. In other words, we compare how much the loss would change non-linearly according to the true dataset gradient, and how much of the per-example gradient noise is present in the dataset with respect to the magnitude of the change. B_{noise} characterizes the dataset at an optimization step. An approach to estimating B_{noise} from a batch in a data parallel setting is described in the comments on data parallelism.

The expected optimal improvement in the loss from the noisy batch gradient is derived as follows.

$$\Delta L_{opt}(B) = \mathbb{E}[L(\theta) - L(\theta - \epsilon_{opt}(B)G_{est})]$$

$$\Delta L_{opt}(B) = L(\theta) - \mathbb{E}[L(\theta - \epsilon_{opt}(B)G_{est})]$$

$$\Delta L_{opt}(B) \approx \epsilon_{opt}(B) |G|^2 - \frac{1}{2} \epsilon_{opt}(B) \frac{|G|^2}{G^T H G + \frac{tr(H\Sigma)}{B}} (G^T H G + \frac{tr(H\Sigma)}{B})$$

$$\Delta L_{opt}(B) \approx \frac{1}{2} \epsilon_{opt}(B) |G|^2$$

$$\Delta L_{opt}(B) \approx \frac{1}{2} \frac{\epsilon_{max}}{1 + \frac{1}{B} B_{noise}} |G|^2$$

$$\Delta L_{opt}(B) \approx \frac{1}{2} \frac{\frac{|G|^2}{G^T H G}}{1 + \frac{1}{B} B_{noise}} |G|^2$$

$$\Delta L_{opt}(B) \approx \frac{1}{2} \frac{\frac{|G|^4}{G^T H G}}{1 + \frac{1}{B} B_{noise}}$$

$$\Delta L_{opt}(B) \approx \frac{\Delta L_{max}}{1 + \frac{1}{B} B_{noise}}, \text{ where } \Delta L_{max} = \frac{1}{2} \frac{|G|^4}{G^T H G}$$

The denominator in the equation for $\Delta L_{opt}(B)$ decreases substantially as B increases when $B \ll B_{noise}$, but not when $B \gg B_{noise}$. Therefore, B_{noise} is a meaningful batch size threshold for the expected improvement of the loss at an optimization step. Batch sizes greater than B_{noise} have diminishing returns.

Setting the Hessian to the identity matrix avoids the overhead of computing the Hessian matrix, and provides B_{simple} , which according to the authors was typically within a small constant from B_{noise} in practice.

Equations 2.11 - 2.12, D.4.

$$\frac{S}{S_{min}} - 1 = \left(\frac{E}{E_{min}} - 1 \right)^{-1}$$

$$B_{crit} = \frac{E_{min}}{S_{min}}$$

B_{noise} is the batch size for the expected optimal improvement of the loss at an optimization step. In contrast, B_{crit} is an optimality measure of a training run at a level of performance. It defines the optimum with respect to the numbers of the executed steps and processed data examples. B_{crit} balances the tradeoff between these numbers at twice the minimal number of steps S_{min} and twice the minimal number of data examples E_{min} for a given level of performance. Otherwise, as the number of steps S grows fast, the number of the processed examples E approaches its minimal number slowly, and vice versa. Therefore, batch sizes other than B_{crit} have diminishing returns with respect to the training speed and data (compute) efficiency.

B_{crit} was measured in the following experiments by fitting data across training runs, performed at fixed batch sizes, to the above equation. Measuring B_{crit} was necessary to empirically connect B_{noise} and B_{simple} to the underlying notion of an optimal training run.

Summary of experiments

Figure 4.

In this and other experiments, training was performed at fixed batch sizes. At an optimization step, B_{simple} was measured according to the procedure described in the comments on data parallelism. At a level of performance, B_{crit} was measured from a fit to the equation 2.11, as described under Figures 6-7.

The relation $B_{crit} \approx B_{simple}$ is shown across tasks and models. For the purposes of Figure 4, B_{simple} was averaged (in addition to the moving average described in the comments on data parallelism). B_{crit} was also averaged over “a full training run”, meaning that a B_{crit} value was obtained from a fit at each level of performance, and the obtained B_{crit} values were averaged. Notably, B_{crit} values were measured from fits to data across training runs, whereas B_{simple} values were measured directly from the gradients of actual training run(s). To be comparable, a B_{crit} value would need to be paired with a B_{simple} value from the training run that resulted in the Pareto front value closest to the B_{crit} value. It is unclear if/how this consideration was implemented in the averaging procedure for Figure 4 and other comparisons between B_{crit} and B_{simple} . Different training runs could be closest to B_{crit} values at different levels of performance. Additionally, the identity of the training run closest to a B_{crit} value could depend on the definition of distance.

Figure 5.

In this and other experiments, learning rates (or learning rate schedules) were tuned for each batch size using a grid search. Figure 5 shows the optimal learning rates across batch sizes in the SVHN classification task with SGD, resembling the relation between B , $\epsilon_{opt}(B)$ and ϵ_{max} in equation 2.6. For each optimal learning rate and the corresponding batch size, the loss improvement measured during training is also shown, resembling the relation between B , $\Delta L_{opt}(B)$ and ΔL_{max} in equation 2.7.

Equation 2.6 was generalized in equation A.3 for other optimizers. The latter was used for centering the grid search for optimal learning rates across different batch sizes. The bounds of the grid search were expanded, if the best learning rate was close to the previous bounds, suggesting that equation 2.6 and its generalized form were less reliable in predicting the optimal learning rate.

Figures 6 - 7.

Across batch sizes and performance levels, the optimal numbers of optimization steps (time efficiency) and data examples (compute efficiency) were measured. A set of batch sizes were chosen. A two-stage grid search was performed to tune the learning rate for each batch size. Although not stated in the paper, it appears that a fixed amount of data parallelism was used for each batch size. After completing the training runs of the grid search, a set of levels of performance were identified. The numbers of steps and examples of the training run that achieved a level of performance “most quickly” were plotted for each batch size. The resulting Pareto fronts were fit to equation 2.11, where each fit was with respect to a level of performance. B_{crit} was then obtained from the fit at each level of performance. The Figures 6-7 show an example of the SVHN classification task with SGD. In Figure 7, the curvature does not appear to change significantly with the decreasing training error on the logarithmic scale, which is consistent with the theoretical prediction that B_{crit} balances the number of optimization steps and the number of examples at twice their minimal numbers for a given level of performance (the minimal numbers increase with the decreasing training error).

Figure 8.

At the time of this study, LSTMs provided an architecture where the number of parameters could be increased by stacking the same building blocks. Figure 8 shows that at a given

training perplexity, the noise scale was roughly the same across different numbers of parameters, although with the increasing number of parameters, lower values of training perplexity could be achieved.

Comments on data parallelism

Data parallelism does not change gradients numerically and provides numerical guarantees. The authors may have used it to avoid sources of numerical variance in the empirical validation.

B_{crit} balances the numbers of steps and processed examples. B_{crit} relates to data parallelism.

Although a batch size $B > B_{crit}$ would provide opportunities for additional data parallelism, such additional parallelism would have diminishing returns, and would require significant additional data and compute, while only slightly reducing the number of steps. Therefore, B_{noise} and B_{simple} optimize data parallelism through the demonstrated relation to B_{crit} .

Data parallelism also provides a convenient setting for estimating $|G|^2$ and $\frac{1}{B}tr(\Sigma)$ during training. The values of the two random variables, defined in equation A.2, were computed from gradients before and after averaging across devices at an optimization step. The expectations of these two random variables were equal to $|G|^2$ and $\frac{1}{B}tr(\Sigma)$, as verified by substituting equation A.2 in equation A.1, and were estimated with moving averages over the optimization steps.

Additional comments

The training runs were performed at fixed batch sizes, the potential value of adaptive batch-sizing was discussed, because B_{crit} and B_{simple} tended to increase with the decreasing training loss.

Although the learning rates needed to be tuned, B_{noise} and B_{simple} are valuable, because they reduce the dimensionality of the hyper-parameter search in training.